Week 4 reading guide and worked examples

Weekly overview

Required learning outcome

By the end of this week, you should be able to calculate partial derivatives, column gradients, Jacobians and Hessians, and apply the multivariable chain rule with dimensionally consistent notation (M6).

Week 4 introduces no new Statistics outcome. Statistics work this week is deliberate consolidation of distributions and moments from S3 and S4. It prepares you to retain the Statistics strand while studying multivariable calculus and before the cumulative Checkpoint A.

Prerequisites and retrieval

You will use one-variable differentiation (M3) and matrix dimensions and multiplication (M4) throughout the new material. The final two questions contain four explicit earlier-outcome retrieval parts:

- a Taylor approximation (M3);
- a matrix–vector product with a dimension check (M4);
- an exponential-distribution probability (S3); and
- expectation and variance under a linear combination (S4).

Before beginning, retrieve the product rule, chain rule and derivatives of $\log x$ and e^x. Also check that you can determine the dimensions of a matrix product before calculating it.

Student materials

1. Use the reading guide for the exact required Osborne sections, the mandatory notation translation and the skip instructions.
2. Study the original institutional notes and worked examples. Cover each solution and reproduce it from a blank page.
3. Attempt all parts of the exercise sheet without the full solutions.
4. Consult the progressive hints only after a serious attempt, and stop as soon as a hint gives you a useful next step.
5. Mark from the complete solutions, recording the first cause of every error by outcome ID.
6. Complete the 24-minute self-test without notes, mark it and schedule a blank-page reattempt of every missed part.
7. Sit Checkpoint A under the stated closed-note conditions when its separate paper is supplied. Checkpoint A is not part of this weekly directory.

The external reading remains on its author's site. Open only the assigned sections. No video, applet, code or specialist software is required.

Suggested study sequence

1. Write the dimensions of a gradient, Jacobian and Hessian at the top of your notes.
2. Complete the 80-minute external reading route, translating every derivative vector into the course convention.
3. Work through the institutional notation notes and reproduce all six examples, checking dimensions before every matrix product.
4. Attempt Questions 1–4, then pause and write the scalar-path and vector-composition chain rules from memory.
5. Attempt Questions 5–8 and finish with the four closed-note retrieval parts in Questions 9–10.
6. Use hints only when needed, then mark all work and record errors under M6, M3, M4, S3 or S4.
7. Take the self-test, update the error log and reattempt at least the first missed step from a blank page.
8. Two or three days later, reattempt every remaining incorrect or incomplete part before using Checkpoint A for cumulative diagnosis.

Workload and required outputs

The times below are provisional planning guidance, not a guarantee, and have not yet been verified by students. They are subject to first-cohort calibration.

Component	Target time	Required output
Required external reading	1 h 20 min	A short original summary of the derivative conventions and chain rules
Institutional notes, worked-example study and reproduction	2 h 40 min	Six examples reproduced with every gradient and Jacobian dimension shown
Week 4 exercise sheet	4 h	All 17 scorable parts attempted and marked
Self-test	24 min	All four self-test parts attempted and marked
Error log and first targeted reattempt	6 min	Error entries by outcome ID and a blank-page reattempt of the first missed step
Checkpoint A	1 h	Separate cumulative paper attempted under its stated conditions
Core total	**9 h 30 min**	Completed Week 4 study and checkpoint cycle

The reading and worked-example component totals 4 hours. The exercise sheet contains 10 numbered questions, 17 scorable parts, 100 marks and a 240-minute target. The self-test, error logging and first targeted reattempt total 30 minutes. The separate 60-minute Checkpoint A brings the complete Week 4 core total to 570 minutes.

Checkpoint A's timed sitting is included in that core total. Its required post-attempt self-marking, diagnosis and spaced reattempt are separate provisional activities: no student-validated minute allowance is claimed, and they are excluded from the published core total pending first-cohort calibration.

End-of-week check

Before moving on, make sure that you can state the input and output dimensions of every map in a composition before multiplying Jacobians. A formula with the right entries in the wrong orientation is not dimensionally correct. Keep the Statistics retrieval entries in the same error log: Week 4 pauses new Statistics teaching but not Statistics retention.

There is no optional or non-examined material in this Week 4 package.

Required reading route

How to use this route

Complete the three named sections below in order. They form one approved resource record and have a combined target time of 80 minutes. Do not treat the surrounding tutorial, its full exercise collection or its interactive elements as assigned.

The source and this course use different derivative-vector conventions. In your notes, translate every scalar-valued derivative into a **column gradient**. For a map from $\mathbb{R}^n$ to $\mathbb{R}^m$, write its Jacobian as an $m \times n$ matrix. The institutional worked examples make that translation explicit.

Required multivariable-calculus reading (M6)

Resource MATH-M6-OSB-REQ; source OSBORNE_MM. In Martin J. Osborne, *Mathematical Methods for Economic Theory*, read:

1. the partial-derivative material only in Section 1.6, "Calculus: many variables";
2. Section 2.1, "Introduction to multivariate calculus"; and
3. Section 2.2, "The chain rule".

Skip: Section 1.6 Exercise 1 and every open-set, closed-set or topology passage; all interactive plots; and Section 2.2 Exercises 10–13. Do not continue into implicit differentiation, optimisation or any later section.

The resource is external-link-only. Use it on the author's site and do not copy its explanations, exercises, diagrams or solutions into shared notes.

Target reading time: 80 minutes.

Required notation translation

After reading, close the source and write the following course conventions from memory:

- if $f : \mathbb{R}^n \to \mathbb{R}$, then $\nabla f(\mathbf{x})$ is $n \times 1$;
- if $\mathbf{f} : \mathbb{R}^n \to \mathbb{R}^m$, then $\mathbf{J_f}(\mathbf{x})$ is $m \times n$;
- the gradient of a scalar-valued function is the transpose of its $1 \times n$ Jacobian;
- the Hessian of a scalar-valued function of n variables is $n \times n$; and
- for $\mathbf{h} = \mathbf{f} \circ \mathbf{g}$, the chain rule uses $\mathbf{J_h} = \mathbf{J_f}(\mathbf{g})\mathbf{J_g}$.

For each line, annotate the dimensions of every factor. If a displayed product is not dimensionally defined, correct its order before doing any calculation.

Statistics consolidation

Week 4 has no new Statistics reading and no new Statistics outcome. Before the worked examples, spend five minutes without notes retrieving:

- the support and survival probability of $X \sim \text{Exponential}(\lambda)$, where $\lambda > 0$ is the rate (S3); and
- the expectation and variance rules for a linear combination, including the covariance term (S4).

This closed-note retrieval is included in the institutional example block, not in the 80-minute external-reading allocation.

Reading output

Produce a short original summary containing one column gradient, one $m \times n$ Jacobian, one Hessian and both forms of the chain rule. State the dimensions beside each object. Do not copy source wording or exercise prompts.

Worked examples

These original institutional notes and examples support M6, with one closing Statistics consolidation example for S3–S4. Allow about 160 minutes to study the notes and reproduce all six examples. For each example, cover the **Solution**, follow the **Method**, and redo the work from a blank page.

Core notation and dimensions

Let $f : \mathbb{R}^n \to \mathbb{R}$ be differentiable. Its gradient is the column vector

$$\nabla f(\mathbf{x}) = \begin{bmatrix} \partial f/\partial x_1 \\ \vdots \\ \partial f/\partial x_n \end{bmatrix} \in \mathbb{R}^n.$$

For a small column displacement $\mathrm{d}\mathbf{x} \in \mathbb{R}^n$, the total differential is

$$\mathrm{d}f = \nabla f(\mathbf{x})^{\mathsf{T}} \mathrm{d}\mathbf{x}.$$

The product has dimensions $(1 \times n)(n \times 1) = 1 \times 1$, as required for a scalar change.

If $\mathbf{F} : \mathbb{R}^n \to \mathbb{R}^m$ has components $F_1, \dots, F_m$, its Jacobian is

$$\mathbf{J}_{\mathbf{F}}(\mathbf{x}) = \left[\frac{\partial F_i}{\partial x_j}\right]_{i=1,\dots,m;\, j=1,\dots,n}.$$

It has one row per output and one column per input, so its dimension is $m \times n$. For a scalar-valued function, the $1 \times n$ Jacobian is the transpose of the $n \times 1$ gradient.

If f has continuous second partial derivatives near the point of interest, its Hessian is

$$\mathbf{H}_f(\mathbf{x}) = \left[\frac{\partial^2 f}{\partial x_i \partial x_j}\right]_{i,j=1,\dots,n}.$$

Under that continuity condition, the mixed partial derivatives agree and the Hessian is symmetric.

Chain-rule forms

For a differentiable path $\mathbf{x} : \mathbb{R} \to \mathbb{R}^n$ and differentiable scalar function $f : \mathbb{R}^n \to \mathbb{R}$,

$$\frac{\mathrm{d}}{\mathrm{d}t} f(\mathbf{x}(t)) = \nabla f(\mathbf{x}(t))^{\mathsf{T}} \mathbf{x}'(t).$$

For differentiable maps

$$\mathbf{g} : \mathbb{R}^p \to \mathbb{R}^n, \qquad \mathbf{F} : \mathbb{R}^n \to \mathbb{R}^m,$$

the composition $\mathbf{H} := \mathbf{F} \circ \mathbf{g}$ maps $\mathbb{R}^p$ to $\mathbb{R}^m$, and

$$\mathbf{J}_{\mathbf{H}}(\mathbf{u}) = \mathbf{J}_{\mathbf{F}}(\mathbf{g}(\mathbf{u}))\mathbf{J}_{\mathbf{g}}(\mathbf{u}).$$

The dimensions are

$$(m \times n)(n \times p) = m \times p.$$

When the outer function is scalar-valued, the equivalent column-gradient form is

$$\nabla(f \circ \mathbf{g})(\mathbf{u}) = \mathbf{J}_{\mathbf{g}}(\mathbf{u})^{\mathsf{T}} \nabla f(\mathbf{g}(\mathbf{u})).$$

Example 1 — a gradient and Hessian (`M6`)

Let

$$f(x, y) = x^2 y + \log y, \qquad (x, y) \in \mathbb{R} \times (0, \infty).$$

Calculate the gradient and Hessian at $(1, 1)$.

Method. Differentiate with respect to one variable while holding the other fixed. State the domain because the logarithm and its derivatives require $y > 0$.

Solution. The first partial derivatives are

$$f_x(x, y) = 2xy, \qquad f_y(x, y) = x^2 + \frac{1}{y}.$$

Thus the column gradient is

$$\nabla f(x, y) = \begin{bmatrix} 2xy \\ x^2 + 1/y \end{bmatrix}, \qquad \nabla f(1, 1) = \begin{bmatrix} 2 \\ 2 \end{bmatrix}.$$

Differentiate the first partial derivatives once more:

$$f_{xx} = 2y, \qquad f_{xy} = 2x, \qquad f_{yx} = 2x, \qquad f_{yy} = -\frac{1}{y^2}.$$

Therefore,

$$\mathbf{H}_f(x, y) = \begin{bmatrix} 2y & 2x \\ 2x & -1/y^2 \end{bmatrix}, \qquad \mathbf{H}_f(1, 1) = \begin{bmatrix} 2 & 2 \\ 2 & -1 \end{bmatrix}.$$

Interpretation. The gradient is 2×1 because there are two inputs and one output. The Hessian is 2×2. The equal off-diagonal entries agree with the mixed-partial symmetry expected from the continuous second derivatives on the stated domain.

Example 2 — the Jacobian of a vector-valued map (`M6`)

Define

$$\mathbf{F} : \mathbb{R}^3 \to \mathbb{R}^2, \qquad \mathbf{F}(x, y, z) = \begin{bmatrix} x^2 y + z \\ x + yz^2 \end{bmatrix}.$$

Calculate the Jacobian at $(1, 2, -1)$.

Method. Put the derivatives of one output component in each row and the derivatives with respect to one input in each column.

Solution. The component derivatives are

$$\nabla F_1(x, y, z) = \begin{bmatrix} 2xy \\ x^2 \\ 1 \end{bmatrix}, \quad \nabla F_2(x, y, z) = \begin{bmatrix} 1 \\ z^2 \\ 2yz \end{bmatrix}.$$

The gradients are columns, but the Jacobian uses their transposes as rows:

$$\mathbf{J}_{\mathbf{F}}(x, y, z) = \begin{bmatrix} 2xy & x^2 & 1 \\ 1 & z^2 & 2yz \end{bmatrix}.$$

At the stated point,

$$\mathbf{J}_{\mathbf{F}}(1, 2, -1) = \begin{bmatrix} 4 & 1 & 1 \\ 1 & 1 & -4 \end{bmatrix}.$$

Interpretation. The map has three inputs and two outputs, so its Jacobian must be 2×3. A 3×2 array containing the same numbers would have the wrong orientation under the course convention.

Example 3 — a scalar path through two variables (M6)

Let

$$f(x, y) = x^2 e^y, \qquad \mathbf{g}(t) = \begin{bmatrix} 1 + t \\ t^2 \end{bmatrix}.$$

Calculate $\mathrm{d}f(\mathbf{g}(t))/\mathrm{d}t$ at $t = 0$ by the multivariable chain rule and verify it by direct substitution.

Method. Multiply the transposed column gradient by the derivative of the column path. Then differentiate the explicit one-variable composition as a check.

Solution. The required derivatives are

$$\nabla f(x, y) = \begin{bmatrix} 2xe^y \\ x^2 e^y \end{bmatrix}, \qquad \mathbf{g}'(t) = \begin{bmatrix} 1 \\ 2t \end{bmatrix}.$$

At $t = 0$, $\mathbf{g}(0) = [1, 0]^{\mathsf{T}}$, so

$$\frac{\mathrm{d}}{\mathrm{d}t} f(\mathbf{g}(t))\Big|_{t=0} = \begin{bmatrix} 2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = 2.$$

For verification, direct substitution gives

$$h(t) := f(\mathbf{g}(t)) = (1 + t)^2 e^{t^2}.$$

Hence

$$h'(t) = 2(1 + t)e^{t^2} + (1 + t)^2 (2t) e^{t^2},$$

and $h'(0) = 2$, as before.

Interpretation. Both coordinates may depend on t, even though the second coordinate happens to have zero derivative at the evaluation point. The dimensional form is $(1 \times 2)(2 \times 1) = 1 \times 1$.

Example 4 — composition of vector-valued maps (M6)

Let

$$\mathbf{g} : \mathbb{R}^2 \to \mathbb{R}^3, \qquad \mathbf{g}(u, v) = \begin{bmatrix} u^2 + v \\ u - v \\ uv \end{bmatrix},$$

and

$$\mathbf{F} : \mathbb{R}^3 \to \mathbb{R}^2, \qquad \mathbf{F}(x, y, z) = \begin{bmatrix} x + 2y - z \\ xy + z^2 \end{bmatrix}.$$

Find the Jacobian of $\mathbf{H} := \mathbf{F} \circ \mathbf{g}$ at $(1, -1)$.

Method. Evaluate the outer Jacobian at the intermediate point $\mathbf{g}(1,-1)$, then multiply it by the inner Jacobian in chain-rule order.

Solution. First,

$$\mathbf{g}(1,-1) = \begin{bmatrix} 0 \\ 2 \\ -1 \end{bmatrix}.$$

The two Jacobians are

$$\mathbf{J}_{\mathbf{g}}(u,v) = \begin{bmatrix} 2u & 1 \\ 1 & -1 \\ v & u \end{bmatrix}$$

and

$$\mathbf{J}_{\mathbf{F}}(x,y,z) = \begin{bmatrix} 1 & 2 & -1 \\ y & x & 2z \end{bmatrix}.$$

At the relevant points,

$$\mathbf{J}_{\mathbf{g}}(1,-1) = \begin{bmatrix} 2 & 1 \\ 1 & -1 \\ -1 & 1 \end{bmatrix}, \qquad \mathbf{J}_{\mathbf{F}}(0,2,-1) = \begin{bmatrix} 1 & 2 & -1 \\ 2 & 0 & -2 \end{bmatrix}.$$

Therefore,

$$\begin{aligned} \mathbf{J}_{\mathbf{H}}(1,-1) &= \mathbf{J}_{\mathbf{F}}(0,2,-1)\mathbf{J}_{\mathbf{g}}(1,-1) \\ &= \begin{bmatrix} 1 & 2 & -1 \\ 2 & 0 & -2 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & -1 \\ -1 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 5 & -2 \\ 6 & 0 \end{bmatrix}. \end{aligned}$$

Interpretation. The dimensions are $(2 \times 3)(3 \times 2) = 2 \times 2$, matching a map from two inputs to two outputs. Reversing the factors gives a 3×3 matrix, not the Jacobian of this composition.

Example 5 — a first-order change in an economic quantity (`M6`)

Let q denote a dimensionless normalised output index and w an input-price scale. Consider the differentiable cost index

$$C(q,w) = w(q+q^2), \qquad q > 0, \quad w > 0.$$

At $(q,w) = (2,3)$, approximate the change in C when $\mathrm{d}q = 1/10$ and $\mathrm{d}w = -1/5$.

Method. Calculate the column gradient and form the total differential. Check that the displaced point remains in the economically relevant positive domain.

Solution. The partial derivatives are

$$C_q(q,w) = w(1+2q), \qquad C_w(q,w) = q+q^2.$$

Thus,

$$\nabla C(2,3) = \begin{bmatrix} 15 \\ 6 \end{bmatrix}.$$

The displaced point is $(2 + 1/10, 3 - 1/5) = (21/10, 14/5)$, whose coordinates are both positive. The first-order change is

$$\begin{aligned} \mathrm{d}C &= \nabla C(2,3)^{\mathsf{T}} \begin{bmatrix} 1/10 \\ -1/5 \end{bmatrix} \\ &= 15\left(\frac{1}{10}\right) + 6\left(-\frac{1}{5}\right) = \frac{3}{2} - \frac{6}{5} = \frac{3}{10}. \end{aligned}$$

Therefore the local approximation predicts an increase of $3/10$ unit in the index.

Interpretation. $C_q(2,3) = 15$ is the local rate with respect to quantity while the input price is held fixed; $C_w(2,3) = 6$ has the analogous holding-quantity-fixed interpretation. The differential combines both changes. It is a first-order approximation for a finite displacement, not an exact equality.

Example 6 — Statistics consolidation (`S3`, `S4`)

Let

$$X \sim \text{Exponential}(2),$$

where the parameter 2 is the rate and the support is $x \geq 0$. Define $Z = 3 - 2X$. Find $\mathbb{P}(X > 1)$, $\mathbb{E}[Z]$ and $\text{Var}(Z)$.

Method. Retrieve the exponential survival probability and its mean and variance under the rate parameterisation. Then apply the linear transformation rules.

Solution. For an exponential random variable with rate $\lambda = 2$,

$$\mathbb{P}(X > x) = e^{-2x}, \qquad x \geq 0,$$

so

$$\mathbb{P}(X > 1) = e^{-2}.$$

Also,

$$\mathbb{E}[X] = \frac{1}{2}, \qquad \text{Var}(X) = \frac{1}{2^2} = \frac{1}{4}.$$

Therefore,

$$\mathbb{E}[Z] = 3 - 2\mathbb{E}[X] = 2$$

and

$$\text{Var}(Z) = (-2)^2 \,\text{Var}(X) = 4\left(\frac{1}{4}\right) = 1.$$

Interpretation. The location shift changes the expectation but not the variance. The multiplier is squared in the variance. This is earlier-outcome consolidation, not a new Week 4 Statistics topic.

Reproduction check

Before starting the exercises, reproduce:

1. the gradient–Jacobian orientation relationship;
2. the dimension line for each form of the chain rule;
3. one gradient, one Jacobian and one Hessian from the examples;
4. the direct check of one composition; and
5. the support and parameterisation used in the Statistics example.

If you cannot predict the dimensions before calculating entries, return to the notation section. Dimension checking is part of the method, not a final cosmetic check.